TASK 4: SMART HOME LIGHTING AUTOMATION

Internship: CodeOrbit IoT Internship

1. Objective

To develop a simple smart home lighting system using Python that switches a light ON or OFF based on motion detection.

2. Tools Used

- Programming Language: Python
- Software: Python IDLE
- Input: User-entered motion status

3. Working Principle

When the user enters 1, the system detects motion and displays “Smart Light: ON”. When the user enters 0, it displays “Smart Light: OFF”. Entering q stops the program.

4. Algorithm

1. Start the program.
2. Get the motion input from the user.
3. If the input is 1, turn the simulated light ON.
4. If the input is 0, turn the simulated light OFF.
5. Repeat until the user enters q.
6. Stop the program.

5. Advantages

- Saves electricity when implemented with suitable sensors.
- Provides convenience.
- Demonstrates basic home automation logic.

6. Conclusion

The Smart Home Lighting Automation program demonstrates how Python can simulate automatic light control based on motion input.

Note: This project is a software simulation and does not control a physical light or sensor.